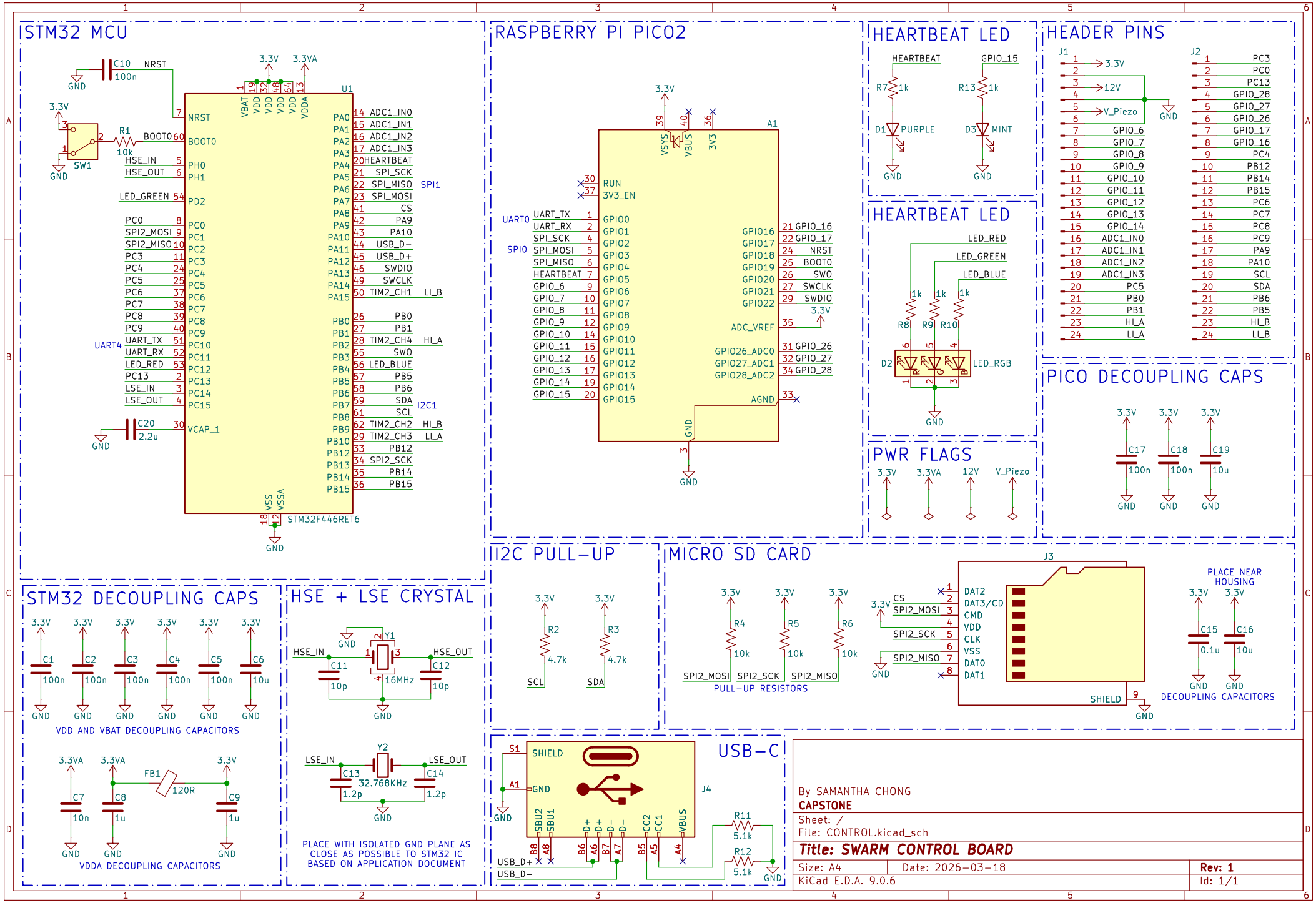
STM32 MCU
U1
STM32F446RET6
NRST
BOOT0
HSE_IN
HSE_OUT
LED_GREEN
UART4
UART_TX
UART_RX
LED_RED
LED_BLUE
SPI1
SPI2_MOSI
SPI2_MISO
SPI2_SCK
I2C1
SDA
SCL
HEARTBEAT
ADC1_IN0
ADC1_IN1
ADC1_IN2
ADC1_IN3
USB_D−
USB_D+
SWDIO
SWCLK
TIM2_CH1
TIM2_CH4
TIM2_CH2
TIM2_CH3
LI_B
HI_A
HI_B
LI_A
RASPBERRY PI PICO2
A1
UART0
SPI0
HEARTBEAT LED
HEARTBEAT
GPIO_15
PURPLE
MINT
HEARTBEAT LED
LED_RED
LED_GREEN
LED_BLUE
LED_RGB
HEADER PINS
J1
J2
PICO DECOUPLING CAPS
PWR FLAGS
3.3V
3.3VA
12V
V_Piezo
I2C PULL−UP
SCL
SDA
MICRO SD CARD
J3
PULL−UP RESISTORS
PLACE NEAR HOUSING
DECOUPLING CAPACITORS
STM32 DECOUPLING CAPS
VDD AND VBAT DECOUPLING CAPACITORS
VDDA DECOUPLING CAPACITORS
HSE + LSE CRYSTAL
16MHz
32.768KHz
PLACE WITH ISOLATED GND PLANE AS CLOSE AS POSSIBLE TO STM32 IC BASED ON APPLICATION DOCUMENT
USB−C
J4
By SAMANTHA CHONG
CAPSTONE
Sheet: /
File: CONTROL.kicad_sch
Title: SWARM CONTROL BOARD
Size: A4
Date: 2026−03−18
Rev: 1
KiCad E.D.A. 9.0.6
Id: 1/1